

IP Addressing — Classes A — E

L / Starting bit
 A 0
 B 10
 :
 110

OCTET

00000001. — Dec

2⁷: 128 2⁴: 16 2⁰: 1
 1 0 1 0 0 0 0 0 0 0
 28 6 4

$$128 + 64 + 32 = 224$$

Req: # Users = 20

"11100000"

What do I have: 255.255.255.0

00000000

Address

Mask -
 NWK HOST

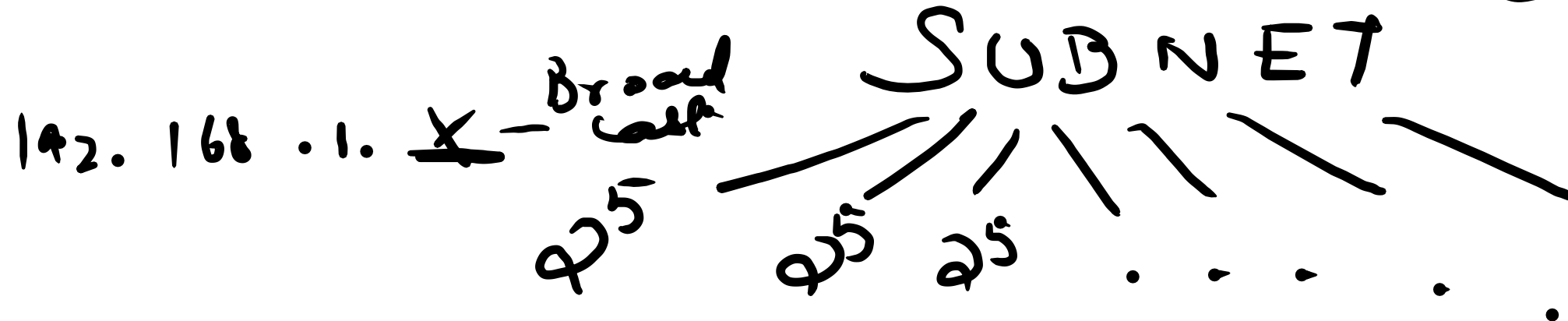
$$2^n \geq 20 \quad 2^5$$

$$n = 5$$

" 192.168.1.0 - Subnet Address, 255.255.255.224 /27

BA

.B*



↑ 32

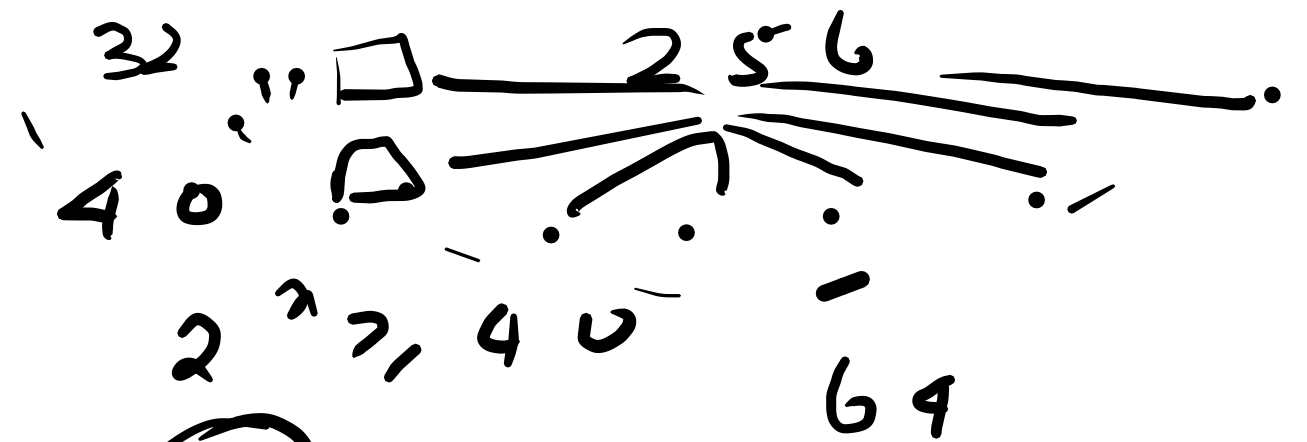
↑ Last

Usable Host Add

Req: 100

$$2^7 > 100$$

255.255.255.0



6

192.168.1.0/24) → Find How bits Want? = (# Users)

b) → Convert Unused 0's into '1's'

"11" 00000' +
2³

c) → Calculate New Subnet Mask

d) → Assign IP address — SA : UHA
BA

S-1

SA 1.0 - 1.31 (BA
1.1-1.30 (UHA)

^{BA} 1.32 — ^{BA} 1.63
1.33-1.62

S-3
1.64-1.95

$$128 + 64 + 32 + 16$$

$$\begin{array}{|c|c|} \hline 0 & 0 & 0 & 0 & 0 & 0 \\ \hline \end{array} \Rightarrow 1111 \boxed{0000} = 255 \cdot 255 \cdot 255 \cdot 255$$

FLSM
VLSM

$$\begin{array}{c} 00000000 \\ 1111 \boxed{000} \end{array}$$

192.168.1.30

SA

2³ - 1

S-1

10-8A

1
2
3
4

5 BA

1.6 -

